Building an Agent-Based Model of T-cell Dynamics

A Step-by-Step Guide for Your BSc Thesis

TYF Vaccination | CD4 (Daan) & CD8 (Tibe)

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What We Will Cover Today

1. What is an ABM and why are we using one?
2. Your data - what it looks like and what it means
3. The biological model - on paper FIRST
4. The 12-step modeling pipeline
5. Calibration, validation, ablation
6. What is wrong with your current code
7. What each of you must deliver

Part 1: What Is an ABM?

And why are we using one?

What Is an Agent-Based Model?

A simulation of INDIVIDUAL entities (agents) following simple rules

POPULATION behavior EMERGES from individual rules

In your project:

- Agent = one T cell
- Rules = activate, choose fate, proliferate, differentiate, die
- Emergent = the population curves you see in the data

Key property: STOCHASTIC

- Same model + same parameters = different result each time
- That's a FEATURE - it gives you uncertainty estimates

ABM vs ODE - When to Use What

Feature	ODE (Compartmental)	ABM (Agent-Based)
Tracks	Pool sizes (continuous)	Individual cells
Random?	No (deterministic)	Yes (stochastic)
Same run twice?	Identical output	Different each time
Variability	Must add manually	Natural from randomness
Per-patient	Hard to individualize	Easy (vary 1-2 params)
Speed	Very fast	Slower (many agents)
Best for	Quick exploration	Final results + uncertainty

Why ABM for YOUR Project?

1. UNCERTAINTY RIBBONS

   You need median +/- IQR from multiple runs

   ABMs give this naturally: run 50 times, take median and IQR

2. PER-PATIENT FITTING

   Each vaccinee is different

   ABM: change 1-2 parameters per person, keep rest fixed

3. STOCHASTIC VARIABILITY

   Is the model's variability SIMILAR to the data's variability?

   If yes: the model captures real biological noise

Use ODE to EXPLORE quickly, ABM for FINAL RESULTS

Part 2: Your Data

48 vaccinees, 4 usable timepoints, 4 subsets

Data Overview - Same Cohort, Two Panels

48 vaccinees (TYF01 - TYF49)

Timepoints: day 1, 22, 43, ~356/365

Days 15 and 29: n = 1 each (IGNORE - not enough for statistics)

Only ~21 vaccinees have 1-year data

What is measured:

%AIM+ = fraction of cells within a subset that activate when stimulated with typhoid antigen in vitro

4 memory subsets: TSCM, TCM, TEM, TEMRA

Values are SMALL: typically 0 to 0.4%

Your model must output in this range!

Daan's Data: CD4 Pattern

Subset	Day 1	Day 22	Day 43	~1 Year
TSCM	0.000	0.218	0.172	0.246-0.286 << PERSISTS!
TCM	0.000	0.381	0.339	0.000-0.178
TEM	0.000	0.000	0.000	0.000 (ALL ZEROS)
TEMRA	0.000	0.072	0.092	0.009-0.034 (declines)

The CD4 story: small but PERSISTENT signal at 1 year in TSCM/TCM

Question: WHY does TSCM persist?
Hypothesis: More cells take the MEMORY route (low q) + slow decay + capacity limit

Tibe's Data: CD8 Pattern

Subset	Day 1	Day 22	Day 43	~1 Year
TSCM	0.000	0.225	0.197	0.000-0.353
TCM	0.000	0.339	0.103	0.000 << SHARP DROP (70%)!
TEM	0.062	0.263	0.222	0.047-0.156
TEMRA	0.000	0.031	0.031	0.005-0.006

The CD8 story: STRONG PEAK at day 22 then SHARP DROP by day 43

Question: WHY the big peak and sharp contraction?
Hypothesis: More cells take the EFFECTOR route (high q) + bigger burst + feedback

Part 3: The Biological Model

Draw it on paper BEFORE you code

The Model Diagram - Know This By Heart

```
Naive Pool (N)
      |
      v  alpha(t) = Gaussian pulse, peak
day 18
  [ACTIVATED]
      |
  +---+---+
  |       |
 MPEC    SLEC          Fate split
(parameter q)
(1-q)    (q)
  |       |
burst   burst          Clonal expansion
  |       |
  v       v
TSCM    TEM            Via daily routing
 TCM    TEMRA
  |       |
SLOW    FAST            Death rates
death   death
```

The Key Insight

Same engine, different bias:

CD4 (Daan):
q ~ 0.3 (more MPEC)
-> explains PERSISTENCE

CD8 (Tibe):
q ~ 0.6 (more SLEC)
-> explains PEAK + DROP

The parameter q IS your scientific hypothesis.
It is NOT just a number.

MPEC vs SLEC - THIS IS YOUR THESIS

MPEC (Memory Precursor)

Will become TSCM or TCM

Lives LONG (months to years)

Responsible for 1-year persistence

Smaller burst (b_MPEC ~ 2)

Think: slow and steady

SLEC (Short-Lived Effector)

Will become TEM or TEMRA

Lives SHORT (days to weeks)

Responsible for big early peak

Bigger burst (b_SLEC ~ 4-5)

Think: fast and furious

Then apply OBSERVATION MODEL:
%AIM+ = cell_count / S
(S = free parameter, per panel)

The Activation Pulse - Get This Right

WRONG (what you both have now)

alpha(t) = alpha_peak if t < T else 0

This is a STEP FUNCTION:

Full activation on day 1

Instant drop to zero on day 24

Biologically WRONG.

Real activation ramps up gradually.

CORRECT (Gaussian pulse)

alpha(t) = alpha_peak *
exp(-0.5 * ((t-18)/7)^2)

This peaks at day 18,

ramps up from day 5,

falls off by day 40,

negligible by day 50.

Shape of activation = shape of response!

Parameters: Fixed vs Free

RULE: You can fit AT MOST 4-5 parameters. You have ~16 data values. 15 free params = overfitting.

Parameter	Value	Meaning	Status
t_peak	18 days	Peak of immune response	FIXED
sigma	7 days	Width of activation pulse	FIXED
mu_TSCM	0.001/day	TSCM death (half-life ~700d)	FIXED
mu_TCM	0.004/day	TCM death (half-life ~170d)	FIXED
mu_TEM	0.015/day	TEM death (half-life ~46d)	FIXED
mu_TEMRA	0.02/day	TEMRA death (half-life ~35d)	FIXED
alpha_peak	0.01-0.3	Activation strength	FREE
S	500-10000	Scale: counts -> %AIM+	FREE
q	0.2-0.8	SLEC fraction (your hypothesis!)	FREE
b or K_mem	varies	Burst size or memory capacity	FREE

The Observation Model: Why You Need It

Your model outputs CELL COUNTS. Your data is %AIM+. These are NOT the same thing.

Layer 1: SIMULATION MODEL

Input: parameters
(q, alpha, death rates, ...)

Output: CELL COUNTS
e.g., TSCM = 45 cells at day 22

This is what you've been building.

Layer 2: OBSERVATION MODEL

Converts counts to %AIM+:

%AIM+(t) = cell_count(t) / S

S = total phenotype pool size
(FREE parameter you calibrate)

THIS IS WHAT YOU ARE MISSING.

Layer 3: LOSS FUNCTION

Compares prediction to data:

This drives calibration.

What S means biologically:

The AIM assay measures: "out of ALL TSCM cells in the sample, what % responded to antigen?"
Your model simulates: "how many antigen-specific TSCM cells were GENERATED by the immune response?"

S bridges these two worlds. It represents the total pool of cells that COULD respond.
It absorbs: total phenotype pool size + assay sensitivity + background correction.

S is NOT a fudge factor. It's a real biological/experimental quantity.
Daan's global_factor = 0.00248 IS an observation model, but implemented as a hack.

CD4 vs CD8: Same Engine, Different Structure

The core daily loop is shared, but these components MUST differ:

Component	CD4 (Daan)	CD8 (Tibe)
Subsets modeled	3 (TSCM, TCM, TEMRA)	4 (TSCM, TCM, TEM, TEMRA)
TEM in obs. model	Fixed at 0 (no signal)	Active, fitted normally
SLEC fraction (q)	~0.35 (MPEC-biased)	~0.6 (SLEC-biased)
Burst ratio	b_SLEC ~ b_MPEC	b_SLEC >> b_MPEC
Memory capacity K_mem	YES (limits growth)	NO (not needed)
Feedback on activation	NO (not needed)	YES (sharpens peak)
Observation model S	S_CD4 (from AIM4)	S_CD8 (from AIM8)
Loss emphasis	Weight PERSISTENCE	Weight CLASSIC-PEAK
Ablation test 1	Remove memory seeding	Remove SLEC (q=0)
Ablation test 2	Remove K_mem (=inf)	Remove feedback (c=0)

These are NOT just different numbers. CD4 has a capacity mechanism CD8 doesn't have. CD8 has a feedback mechanism CD4 doesn't have. The data tells us this.

The Observation Model Differs: AIM4 vs AIM8

Daan: AIM4 Panel (CD4+ T cells)

What the assay measures:
Among all CD4+ memory T cells,
what % express activation markers
after TYF antigen stimulation?

Observation model:
%AIM+_TSCM = TSCM_count / S_CD4
%AIM+_TCM = TCM_count / S_CD4
%AIM+_TEM = 0 (hardcoded)
%AIM+_TEMRA = TEMRA_count / S_CD4

S_CD4 = total CD4+ pool size
(a FREE parameter)

TEM is forced to zero because
the data shows NO CD4 TEM response.

Tibe: AIM8 Panel (CD8+ T cells)

What the assay measures:
Among all CD8+ memory T cells,
what % express activation markers
after TYF antigen stimulation?

Observation model:
%AIM+_TSCM = TSCM_count / S_CD8
%AIM+_TCM = TCM_count / S_CD8
%AIM+_TEM = TEM_count / S_CD8
%AIM+_TEMRA = TEMRA_count / S_CD8

S_CD8 = total CD8+ pool size
(a FREE parameter)

All 4 subsets are active.
TEM shows the clearest peak-and-drop.

S_CD4 and S_CD8 are INDEPENDENT parameters - different assays, different total pools.

Fixed Parameters: Values and Literature Sources

Parameter	Symbol	Value	Unit	Source Paper
Naive death rate	mu_N	0.0003	/day	Borghans et al. 2018, Immunol Rev
TSCM death rate	mu_TSCM	0.0002	/day	Costa del Amo et al. 2018, PLOS Biol
TCM death rate	mu_TCM	0.004	/day	Macallan et al. 2004, JEM
TEM death rate	mu_TEM	0.01	/day	Macallan et al. 2004, JEM
TEMRA death rate	mu_TEMRA	0.02	/day	Macallan et al. 2004, JEM
MPEC death rate	mu_MPEC	0.02	/day	Joshi et al. 2007, Immunity
SLEC death rate	mu_SLEC	0.05	/day	Joshi et al. 2007, Immunity
MPEC->TSCM route	f_TSCM	0.03	/day	Farber et al. 2014, Nat Rev Immunol
MPEC->TCM route	f_TCM	0.05	/day	Farber et al. 2014, Nat Rev Immunol
SLEC->TEM route	f_TEM	0.06	/day	Farber et al. 2014, Nat Rev Immunol
SLEC->TEMRA route	f_TEMRA	0.02	/day	Farber et al. 2014, Nat Rev Immunol

Death rates derived from half-lives using: mu = ln(2) / half-life_in_days

CD4 vs CD8: Different Fixed + Free Parameters

Daan - CD4 Parameters

FIXED (from literature):

t_peak = 18 days (Akondy 2017)
sigma = 7 days (Akondy 2017)
q = 0.35 MPEC-biased (Kaech & Cui 2012)
b_SLEC = b_MPEC (tied) (CD4: similar burst)
+ all shared death & routing rates

FREE (Optuna searches these):

alpha_peak [0.01-0.3] activation strength
b_MPEC [1-5] burst size
K_mem [50-500] memory capacity
S_CD4 [500-10000] observation scale

4 free / 12 data values = ratio 3.0

Tibe - CD8 Parameters

FIXED (from literature):

t_peak = 14 days (Akondy 2017)
sigma = 5 days (Akondy 2017)
q = 0.6 SLEC-biased (Joshi 2007)
b_MPEC = 2 (fixed) (Kaech & Ahmed 2001)
+ all shared death & routing rates

FREE (optimizer searches these):

alpha_peak [0.01-0.3] activation strength
b_SLEC [3-8] SLEC burst size
feedback_c [0.001-0.01] feedback strength
S_CD8 [500-10000] observation scale

4 free / 16 data values = ratio 4.0

Part 4: The 12-Step Pipeline

You are NOT allowed to skip steps

The 12-Step Modeling Pipeline

1 QUESTION
State it in one sentence

2 EDA
Know your data

3 HYPOTHESIS
What mechanism explains the pattern?

4 MODEL DESIGN
Draw on paper, write equations

5 IMPLEMENTATION
Code the model

6 SANITY CHECKS
Extreme-case verification

7 CALIBRATION
Find params that match data

8 VALIDATION
Does it generalize?

9 SENSITIVITY
Which params matter most?

10 ABLATION
Is each mechanism necessary?

11 PER-PATIENT
Explain individual differences

12 INTERPRETATION
What does it tell us?

Step 6: Sanity Checks - DO NOT SKIP

Check	What You Do	Expected Result
1	Set alpha_peak = 0	All pools stay at zero
2	Set q = 0 (all MPEC)	TEM/TEMRA get NO signal
3	Set q = 1 (all SLEC)	TSCM/TCM get NO signal
4	Set all death rates = 0	Pools grow monotonically
5	Run 365 days	No negative values anywhere
6	Plot raw output	Looks like T-cell dynamics?

If ANY check fails, you have a BUG. Fix it before proceeding to calibration.

Part 5: Calibration

Finding parameters that match your data

Calibration: What It IS vs What It ISN'T

Calibration IS:

Systematic search for parameter values
that minimize a defined loss function

Using multiple seeds per evaluation (50+)

With at most 4-5 free parameters

Producing reproducible best-fit values

Calibration is NOT:

Moving sliders until it 'looks right'

Fitting all 15 parameters at once
(= overfitting, proves nothing)

Getting a perfect fit
(perfect fit + many params = meaningless)

A single stochastic run per evaluation

The Loss Function

```
Loss = RMSE(model_medians, data_medians)
     + w1 * |model_responder%  - data_responder%|
     + w2 * |model_classic_peak% - data_classic_peak%|
     + w3 * |model_persistence% - data_persistence%|
```

The weights encode what you CARE about:

Daan (CD4): w3 high - weight PERSISTENCE more
Tibe (CD8): w2 high - weight CLASSIC-PEAK more

CRITICAL: Each loss evaluation = run 50 seeds, use MEDIAN output
A single stochastic run is meaningless for calibration!

Daan: you have Optuna already. Good. But fix the model first,
and change from 1 seed per trial to 50.

SSE compares PREDICTED %AIM+ (from observation model) to DATA %AIM+, not raw cell counts!

After Calibration: 3 Mandatory Plots

PLOT 1: Model vs Data Medians

- x = days, y = %AIM+
- Model curve (median of 50 seeds) with IQR ribbon
- Data points with IQR error bars, one panel per subset

PLOT 2: Cohort Metrics Comparison (bar chart)

- Side-by-side: data vs model
- Bars for: Responder %, Classic-peak %, Persistence %

PLOT 3: Residuals

- model value - data value at each timepoint/subset
- Should be small and random (no systematic pattern)
- Systematic bias = something is wrong in your model

Part 6: Ablation Tests

Prove that your mechanism is NECESSARY

Ablation = Remove One Mechanism, See What Breaks

Analogy: your car goes fast. You think the turbo matters.

Proof: remove the turbo, keep everything else, show it's slow now.

RULES:

1. Change ONE thing at a time
2. Do NOT refit after ablation (keep best-fit params!)
3. Compare BOTH curves AND metrics
4. Write 3-4 sentences explaining the result

If you refit after ablation, you defeat the entire purpose.

The point is to show the model BREAKS without the mechanism.

Your Ablation Tests

Daan (CD4) - 2 Tests

Test 1: No Memory Seeding

Set f_TSCM = 0, f_TCM = 0

Expected: 1-year tail DISAPPEARS

Proves: memory route is necessary

Test 2: No Memory Capacity

Set K_mem = infinity

Expected: 1-year values TOO HIGH

Proves: capacity is necessary

Tibe (CD8) - 2 Tests

Test 1: No SLEC Branch

Set q = 0 (all cells go MPEC)

Expected: early peak FLATTENS

Proves: SLEC is necessary for peak

Test 2: No Feedback

Set feedback_c = 0

Expected: peak TOO BROAD/HIGH

Proves: feedback sharpens the peak

Part 7: What Is Wrong With Your Current Code

Honest assessment of your latest submissions

Daan: Issues in ABM_V4

Issue	Status	Impact
No MPEC/SLEC intermediate stage	MISSING	Cannot test thesis hypothesis
Step function activation (not Gaussian)	WRONG	Wrong curve shapes
No memory capacity (K_mem)	MISSING	Cannot run ablation test 2
14 free parameters in Optuna	WRONG	Overfitting (absurd best-fit values)
TSCM half-life = 9,800 years	ABSURD	Not biology
ABM_variance_V4.py not in zip	BROKEN	Code cannot run
1 seed per Optuna trial	WRONG	Noisy, unreliable loss
Scale mismatch (0-100 vs 0-0.4)	BUG	Wrong comparison with data

Tibe: Issues in Latest Code

Issue	Status	Impact
Dead cells proliferate (no continue after death)	BUG	Inflates all cell counts
Dead cells never removed from list	BUG	Simulation gets slower every day
Cell list stored as reference, not copy	BUG	Historical data corrupted
slec_fraction = 0.2 (should be 0.6)	WRONG	Model behaves like CD4, not CD8
No feedback mechanism	MISSING	Cannot run ablation test 2
No burst at activation (b_SLEC > b_MPEC)	MISSING	Peak not strong enough
No systematic calibration	MISSING	Only manual slider tuning
ren_S = 0.03 (10x too high biologically)	WRONG	Unrealistic renewal rate

Part 8: What You Must Deliver

Figures, tables, and text

What To Fix FIRST (Priority Order)

Daan - Fix List

1. Add MPEC/SLEC fate split (the core!)
2. Replace step function with Gaussian pulse
3. Add memory capacity (K_mem)
4. Fix scaling (remove global_factor)
5. Reduce to 4-5 free parameters
6. Include ALL files in submission
7. Add seeds (50 per Optuna trial)
8. Run sanity checks
9. Calibrate properly
10. Ablation tests

Tibe - Fix List

1. Fix proliferation-after-death (add continue)
2. Remove dead cells periodically
3. Remove ListCells from getData
4. Add feedback mechanism
5. Add burst at activation
6. Set slec_fraction = 0.6
7. Fix ren_S to 0.002 (not 0.03)
8. Add seeds, implement loss function
9. Calibrate systematically
10. Ablation tests

Timeline: 8 Weeks to Submission

Week	Task	Deliverable
1	Fix bugs + add missing mechanisms	Working model code
2	Sanity checks (all 6 must pass)	Sanity check report
3	Calibration (loss function + search)	Best-fit parameters + table
4	Fit quality plots + sensitivity	3 mandatory plots + tornado
5	Per-patient fitting (8 vaccinees)	8 overlay figures
6	Ablation tests (2 per student)	Ablation figures + text
7	Write-up (methods + results)	Draft thesis sections
8	Polish figures + final check	Submission-ready package

References: Primary Sources

[1] Joshi NS et al. (2007) Immunity 27(2):281-295
Inflammation directs MPEC and SLEC CD8+ T cell fates via graded T-bet expression.
Used for: MPEC/SLEC paradigm, q values, intermediate state death rates

[2] Kaech SM, Cui W. (2012) Nat Rev Immunol 12:749-761
Transcriptional control of effector and memory CD8+ T cell differentiation.
Used for: MPEC/SLEC framework review, CD4 vs CD8 q rationale

[3] Macallan DC et al. (2004) J Exp Med 200(2):255-260
Rapid turnover of effector-memory CD4+ T cells in healthy humans.
Used for: TCM, TEM, TEMRA death/proliferation rates (deuterium labeling)

[4] Costa del Amo P et al. (2018) PLOS Biol 16(6):e2005523
Human TSCM cell dynamics in vivo are shaping the T cell compartment.
Used for: TSCM half-life = 8.7 years (slow fraction), self-renewal rate

[5] Borghans JAM et al. (2018) Immunol Rev 285:119-134
Current best estimates for the average lifespans of mouse and human lymphocytes.
Used for: Naive T-cell lifespan (~6-9 years), meta-analysis of deuterium studies

[6] Akondy RS et al. (2017) Nature 552:362-367
Origin and differentiation of human memory CD8 T cells after vaccination.
Used for: Activation peak timing (day 11-14 for CD8), memory formation timeline

References: Primary Sources (continued)

[7] Fuertes Marraco SA et al. (2015) Sci Transl Med 7(282):282ra48
Long-lasting stem cell-like memory CD8+ T cells upon yellow fever vaccination.
Used for: TSCM persistence up to 25 years post-vaccination

[8] Farber DL et al. (2014) Nat Rev Immunol 14:24-35
Human memory T cell generation, compartmentalization, and homeostasis.
Used for: Memory differentiation hierarchy, routing fraction rationale

[9] Surh CD, Sprent J. (2008) Immunity 29(6):848-862
Homeostasis of naive and memory T cells.
Used for: Memory niche concept, IL-7/IL-15 competition, K_mem rationale

[10] Gattinoni L et al. (2011) Nat Med 17:1290-1297
A human memory T cell subset with stem cell-like properties.
Used for: TSCM identification, multipotency, self-renewal capacity

[11] Kaech SM, Ahmed R. (2001) Nat Immunol 2:415-422
Memory CD8+ T cell differentiation: initial antigen encounter triggers a developmental program in naive cells.
Used for: Burst size estimates (10,000-fold expansion per naive precursor)

[12] Westera L et al. (2013) Blood 122(13):2205-2212
Closing the gap between T-cell lifespan estimates.
Used for: Reconciling deuterium labeling half-life measurement methods

[13] Wherry EJ, Ahmed R. (2004) J Virol 78(11):5535-5545
Memory CD8 T-cell differentiation during viral infection.
Used for: Effector-mediated contraction, feedback mechanism concept

Final Deliverables Checklist

FIGURES:

[ ] Median +/- IQR: model vs data (per subset, over time)

[ ] Bar chart: responder / classic-peak / persistence (data vs model)

[ ] 8 per-patient overlays with uncertainty ribbons

[ ] 2 ablation figures with before/after metrics

[ ] Sensitivity tornado plot

TABLES:

[ ] Parameter table (name, value, fixed/free, source)

[ ] Best-fit parameter values

[ ] Ablation metrics comparison

TEXT:

[ ] 1 paragraph: question + hypothesis

Remember This

Daan:
"Show that a MEMORY-biased route makes TSCM/TCM persist at 1 year.
Prove it by turning memory off."

Tibe:
"Show that an EFFECTOR-biased route with a big burst makes the peak and drop.
Prove it by turning SLEC off."

Draw the model on paper.
Fix the biology.
Then calibrate.
Then ablate.

In that order. No shortcuts.